Machine Learning Approach to Cardiovascular Disease (CVD) Prediction

Fulvio Favilla Filho — University of São Paulo (USP)

Mentors: Alysson Oliveira, Mauro Marques — LexisNexis Risk Solutions

Abstract

Cardiovascular diseases are the leading cause of global mortality, presenting a significant challenge to public health.

The main objective of this study is to create predictive models through HPCC bundles Logistic Regression and Learning Trees, comparing both models based on specific metrics to select the most appropriate one for prediction. Through the analysis and interpretation of simple clinical data, the model seeks to provide accurate predictions to identify patients at risk of developing CVD, enabling preventive interventions and more effective treatments.

Data Collection

The dataset used for model development contains 70,000 patient records, encompassing various types of features: *objective*, *examination*, and *subjective*.

In addition to the creation of predictive models, an analysis of the importance and correlation between each feature during CVD prediction is conducted, allowing for a better understanding of the relevance of different characteristics in cardiovascular health.

Resources:

1. kaggle.com/sulianova/cardiovascular-disease-dataset
2. github.com/fulviofavilla/cvd-detection-ml

Data Preparation

In the data preparation stage, noise was removed by addressing outliers in features like **blood pressure**, **weight**, and **height**. Class balancing for the target variable **cardio** was performed to mitigate class imbalance and avoid prevalence using undersampling techniques.

Additionally, the **age** field, initially in days, was converted to years, and the **weight** was transformed from float to integer, as the precision of one decimal place was not relevant for model training. Categorical variables like **gender** were passed to the nominal fields hyperparameter for the Learning Trees model.

After this process, a total of 60,000 valid records were obtained from the initial dataset for training the models. These preprocessing steps enhanced data quality and facilitated effective analysis and modeling.

Implementation

For model implementation, the Data Patterns bundle was employed to identify the Best Record for the dataset. Subsequently, Logistic Regression and Random Forest models were developed to compare the performance of these models in CVD prediction.

Logistic Regression

Initially, training using Binomial Logistic Regression involved all **11** independent variables. Subsequently, a Correlation Matrix analysis was conducted through the CORRELATION function, calculated using *Pearson's Product Moment Correlation Coefficient*.

This process revealed that equivalent accuracy in the Logistic Regression model could be achieved by using only **6** independent variables — those exhibiting significant correlation with the target variable **cardio**.

This observation led to a lower **Akaike Information Criterion (AIC)** during the selection of independent variables, indicating a less complex and more optimized model configuration.

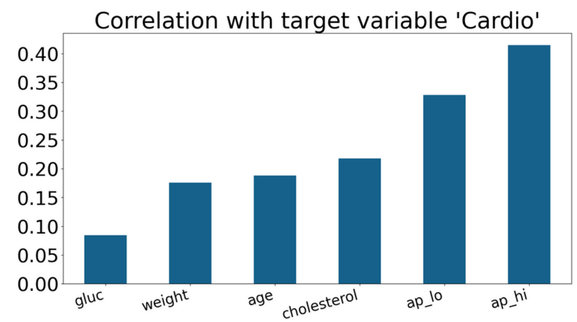

Figure 1: Features with significative correlation with target variable

Random Forest

For the RandomForest model, the Learning Trees bundle using the Classification Forest module was utilized. Initially, default hyperparameter settings were employed, followed by fine-tuning and specifying categorical fields as nominal fields during model adjustments for enhanced performance.

Results

	Binomial Logistic Regression	
Metrics	Before	After
Accuracy	**71.8%**	**72.3%**
Precision	**73.7%**	**75.9%**
Recall	**67.9%**	**66.0%**
AIC	**16318**	**13611**

Table 1: Logistic Regression metrics before and after optimization

	Classification Forest	
Metrics	Before	After
Accuracy	**71.6%**	**79.1%**
Precision	**71.8%**	**82.3%**
Recall	**70.8%**	**74.5%**

Table 2: Random Forest metrics before and after optimization

	RF Model - Feature Importance	
Feature Ranking	Before	After
weight	**29.8%**	**24.9 %**
height	**18.9%**	**23.1 %**
age	**18.2%**	**18.1 %**
ap_hi	**14.9%**	**15.5 %**
ap_lo	**8.0%**	**8.6 %**

Table 3: Feature importance of the RF model before and after optimization

Conclusion

In conclusion, the outcomes clearly indicate that the **Random Forest** model is more suitable for cardiovascular disease prediction within the analyzed dataset when compared to **Logistic Regression**.

Nevertheless, it remains evident for both models that the preparation of data and the subsequent optimization significantly contribute to achieving superior machine learning performance. These methodologies are not only valuable for prospective investigations within this domain but also hold potential for broader application across diverse datasets.